

Keshav Rajput

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EDUCATION

New York University, Tandon School of Engineering

MS, Computer Science, May 2026. GPA 3.67/4.0

IIIT Delhi

B.Tech, Computer Science and Applied Mathematics; minor in Economics, 2020 - 2024. CGPA 8.19/10

Coursework: Stochastic Processes, Statistical Inference, Convex Optimization, Game Theory, Econometrics, Valuation and Portfolio Management, Machine Learning

PROJECTS

Non-stationary bandits | Python, numpy github.com/happyc0der/multi_armed_bandit_algorithms, 2026

- Five non-stationary bandit algorithms (TS-GE, AdSwitch, M-UCB, UCB1, epsilon-greedy) behind one interface with a regret benchmark.
- Fixed a reversed Beta update in TS-GE: regret 21,860 to 174. Showed forced probing is 76% of its total regret.
- Made AdSwitch 59x faster (218 s to 3.7 s) with bit-identical output.

Kaggle S6E9, EV purchases | Python, LightGBM, XGBoost, github.com/happyc0der/kaggle-s6e9-ev-purchases, 2026
CatBoost, PyTorch, Optuna

- Kaggle Playground S6E9 (EV purchases): out-of-fold AUC 0.9465, public leaderboard 0.94649 against a leader at 0.94674.
- Nested target encoding feeding LightGBM, XGBoost, CatBoost and an MLP; 669k training rows.
- 190+ logged experiments compared with paired DeLong tests; showed the remaining gap is label noise.

Snakes and Ladders as a Markov chain | Python, numpy github.com/happyc0der/SPA_Project, 2022

- Snakes and Ladders as a 101-state absorbing Markov chain; expected game length 39.6 turns from the fundamental matrix.
- Cross-checked by Monte Carlo; a die-size sweep finds a 15-sided die is fastest (25.8 turns). 47 tests.

Subgradient descent with momentum | Python, PyTorch github.com/happyc0der/AOMML, 2024

- Stochastic subgradient descent with and without momentum, written as PyTorch optimizers, on LASSO and an MLP.
- Matches scikit-learn's LASSO optimum to $7.4e-5$; momentum's apparent speedup drops to 1.05x once step size is matched. 89 tests.

COMPETITIONS

- **Biohub Cell Tracking, Kaggle**, 2026. Public leaderboard 0.952, about top 1.2% of 3,594 teams.
- **Kaggle Playground S6E9**, 2026. AUC 0.9465, 0.00025 behind the leader.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash

ML: PyTorch, scikit-learn, LightGBM, XGBoost, CatBoost, Optuna, NumPy, pandas, polars

Data: PySpark, Databricks, SQL, PostgreSQL, SQLite, pandas, Streamlit, Altair, matplotlib

Tools: Git, GitHub Actions, Docker, AWS (CDK, App Runner, DynamoDB), Linux, LaTeX

ACTIVITIES

- Events head, Finnexia (finance club), IIIT Delhi; organised "The Critical Bid"
- Attended the first International Quantum Communication Conclave, New Delhi, 2023